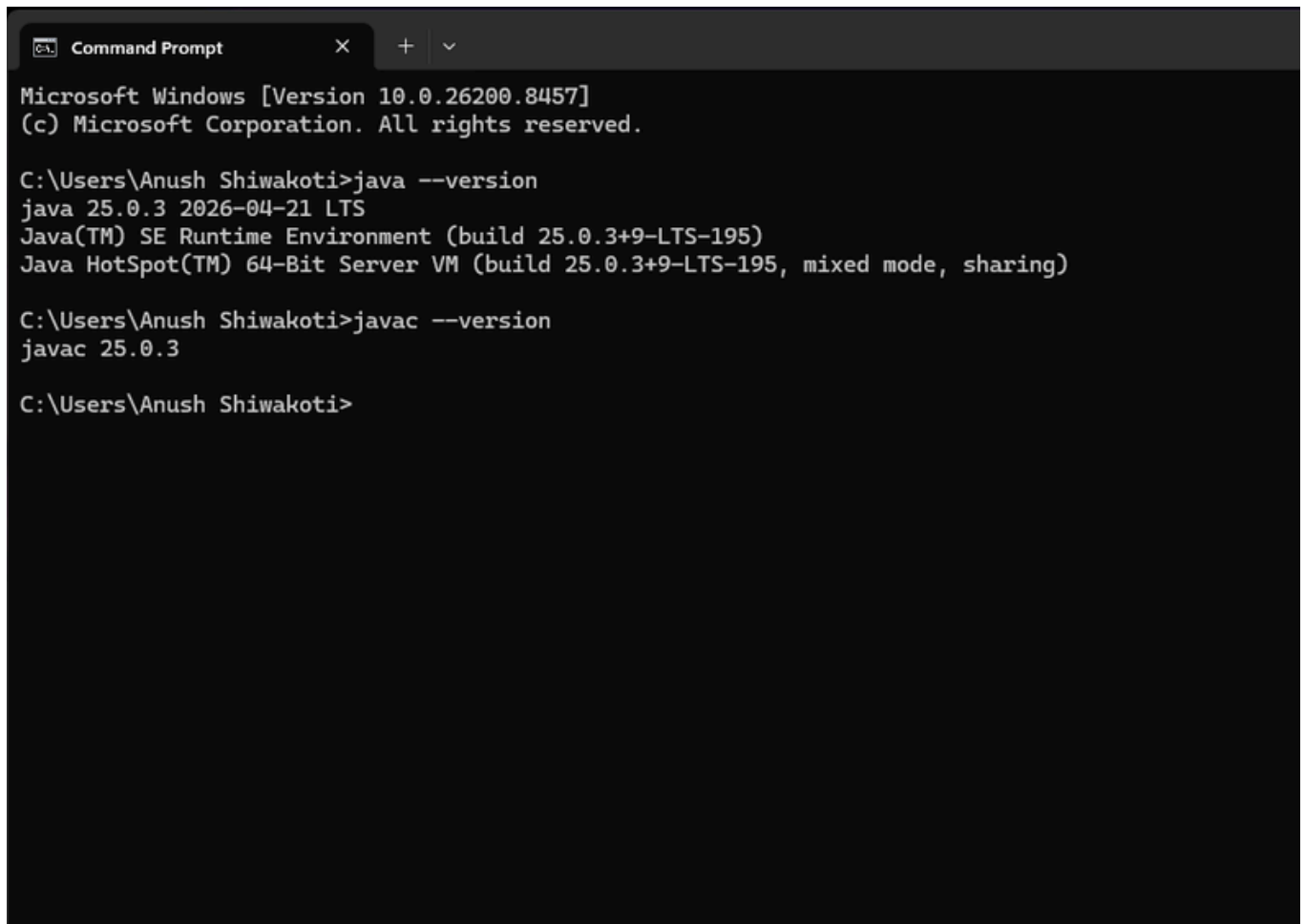


OOP TUTORIAL 1

A screenshot of a Windows Command Prompt window. The title bar shows 'Command Prompt' with standard window controls. The text inside the window shows the Windows version and copyright information, followed by two commands: 'java --version' and 'javac --version'. The output for 'java --version' shows 'java 25.0.3 2026-04-21 LTS' and details about the Java SE Runtime Environment and HotSpot VM. The output for 'javac --version' shows 'javac 25.0.3'. The prompt is currently at 'C:\Users\Anush Shiwakoti>'.

```
Microsoft Windows [Version 10.0.26200.8457]
(c) Microsoft Corporation. All rights reserved.

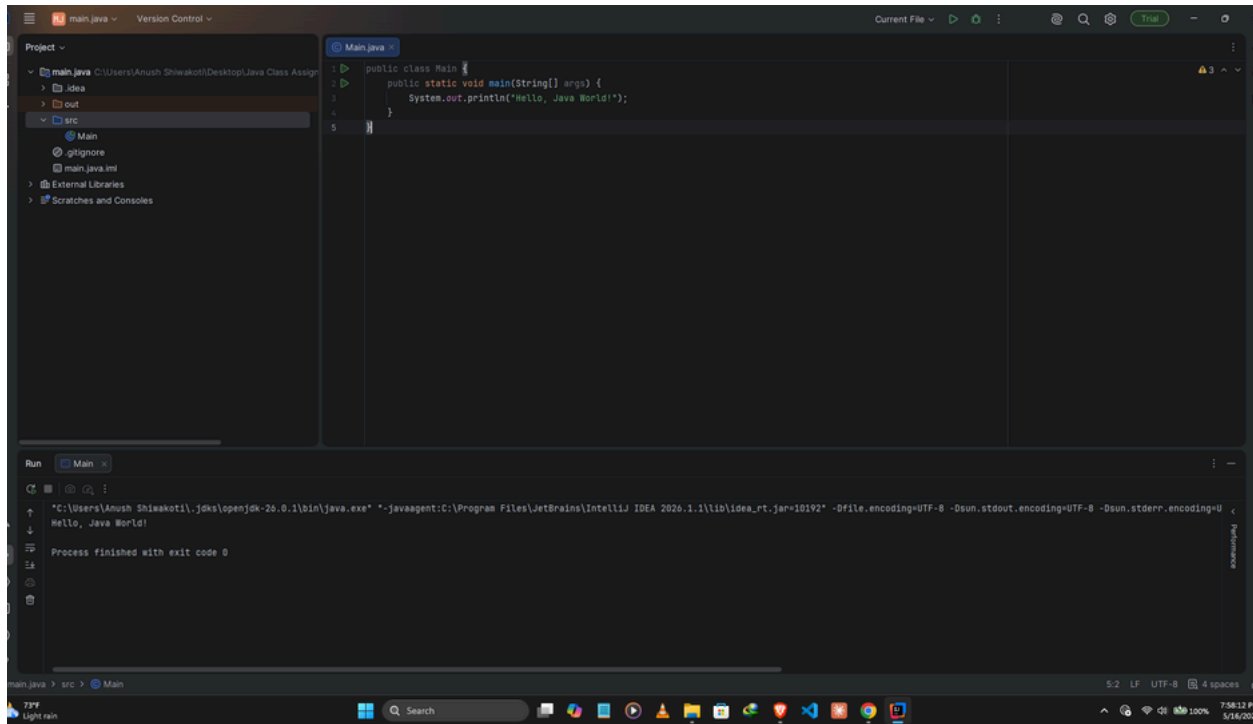
C:\Users\Anush Shiwakoti>java --version
java 25.0.3 2026-04-21 LTS
Java(TM) SE Runtime Environment (build 25.0.3+9-LTS-195)
Java HotSpot(TM) 64-Bit Server VM (build 25.0.3+9-LTS-195, mixed mode, sharing)

C:\Users\Anush Shiwakoti>javac --version
javac 25.0.3

C:\Users\Anush Shiwakoti>
```

Screenshot 1: Java Installation Verification

This screenshot displays the successful installation of the Oracle JDK on my local machine. By running the `java --version` and `javac --version` commands in the terminal, I have verified that the system environment variables are correctly configured to run and compile Java applications.



Screenshot 2: IDE Setup and Project Workspace

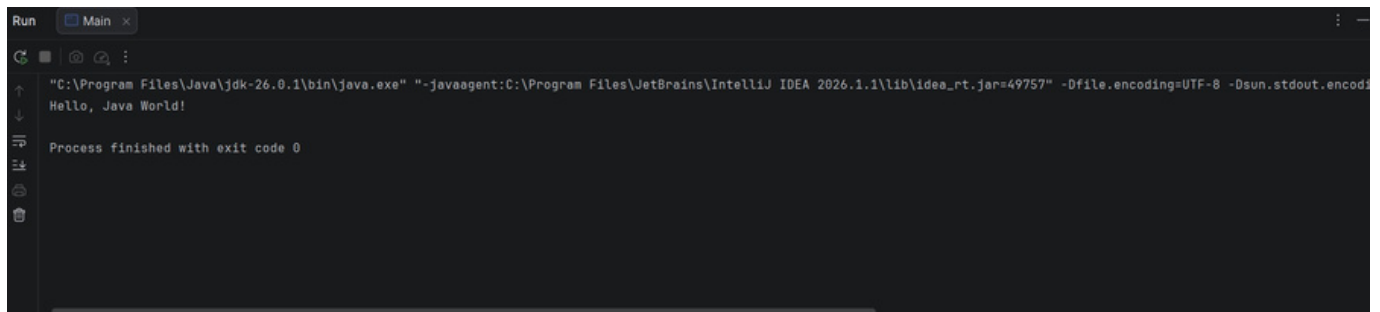
This screenshot demonstrates that IntelliJ IDEACommunity Edition has been successfully installed and configured with the Java DevelopmentKit (JDK). The workspace shows the newly created project for the tutorial tasks, displayingtheproject directory structure and the initialized classes readyfordevelopment

Hello World Execution

```
Hello world Program
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello, Java World!");
    }
}
```

Handwritten Code

OUTPUT



```
Run Main x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=49757" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8
Hello, Java World!
Process finished with exit code 0
```

Variables and Data Types

```
Variable & Data Types Program
public class Main {
    public static void main (String[] args)
    {
        int age = 19;
        double height = 5.8;
        boolean student = true;
        char grade = 'A';
        String name = "Anush"

        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
        System.out.println("Student: " + student);
        System.out.println("Grade: " + grade);
    }
}
```

HANDWRITTEN CODE

OUTPUT

```
Run  datatypes x
C:\Program Files\Java\jdk-26.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=
Name: Anush
Age: 19
Height: 5.8
Student: true
Grade: A
Process finished with exit code 0
```


Operators Program

```
Operators Program
public class Operators {
    public static void main (String[] args)
    {
        int a=10;
        int b=5;

        System.out.println("Addition:" + (a+b));
        System.out.println("Subtraction:" + (a-b));
        System.out.println("Multiply:" + (a*b));
        System.out.println("Division:" + (a/b));
        System.out.println("Modulus:" + (a%b));

        System.out.println(a>b);
        System.out.println(a<b);
        System.out.println(a==b);
        System.out.println(a!=b);
        System.out.println((a>b)&&(b>0));
        System.out.println((a<b)|| (b>0));
        System.out.println(! (a<b));
    }
}
```

HANDWRITTEN CODE

OUTPUT

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program F
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2
Modulus: 0
true
false
false
true
true
true
true
```

User Input

```
User Input Program
import java.util.Scanner;
public class UserInput {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first number: ");
        int a = sc.nextInt();

        System.out.print("Enter second number: ");
        int b = sc.nextInt();

        int sum = a + b;

        System.out.println("Sum = " + sum);
    }
}
```

HANDWRITTEN CODE

OUTPUT

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:
Enter first number: 1
Enter second number: 2
Sum = 3

Process finished with exit code 0
|
```